SCP-55 DOCK

Rev.A / functional schematic

USB / DIN MIDI -> RP2354B -> SCP-55 -> I2S DAC -> LINE OUT

4-layer PCB / 73 x 115 mm / RP2354B with 2 MB internal flash

Read left to right within each functional block. Identical global labels connect across pages.

USB-C input and 3.3 V supply

02

ファイル: scp55-usb-power.kicad_sch

MCU / card bus and peripherals

03

ファイル: scp55-mcu-signals.kicad_sch

MCU / clock, boot and power

04

ファイル: scp55-mcu-support.kicad_sch

Card bus / address and data translation

05

ファイル: scp55-bus-address-data.kicad_sch

Card bus / control and return signals

06

ファイル: scp55-bus-control.kicad_sch

PC Card socket / pinout and fixed options

07

ファイル: scp55-card-socket.kicad_sch

Card power / protection and interlock

08

ファイル: scp55-card-control.kicad_sch

Card detection / Schmitt inputs

09

ファイル: scp55-card-detect.kicad_sch

DIN MIDI input / activity indicators

10

ファイル: scp55-midi-in.kicad_sch

Audio output / PCM5102A DAC

11

ファイル: scp55-dac.kicad_sch

Test points / signal reference

12

ファイル: scp55-debug.kicad_sch

Electrical design: Rev.A production schematic, 2026-09-30. Layout edition: 2026-11-08.

Hardware: CERN-OHL-P-2.0 / Copyright 2026 Shuichi Takano

CERN-OHL-P-2.0

Sheet: /
File: scp55-reva.kicad_sch

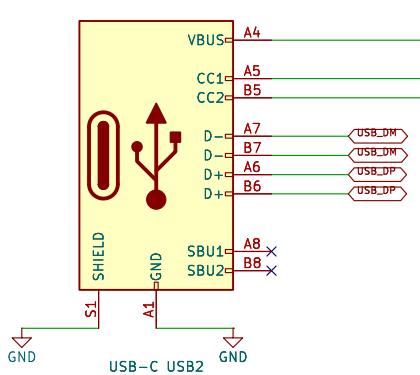
Title: SCP-55 Dock / schematic guide

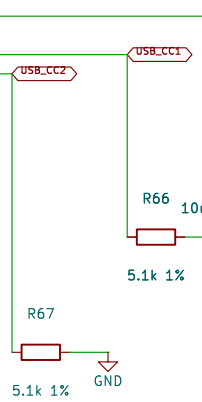
Size: A3	Date:	Rev: A / readable
KiCad E.D.A. 9.0.6		Id: 1/12

USB-C INPUT AND 3.3 V SUPPLY

USB2 device / 5 V sink. USB data continues to the MCU on page 4.

01 / USB-C + 5 V input

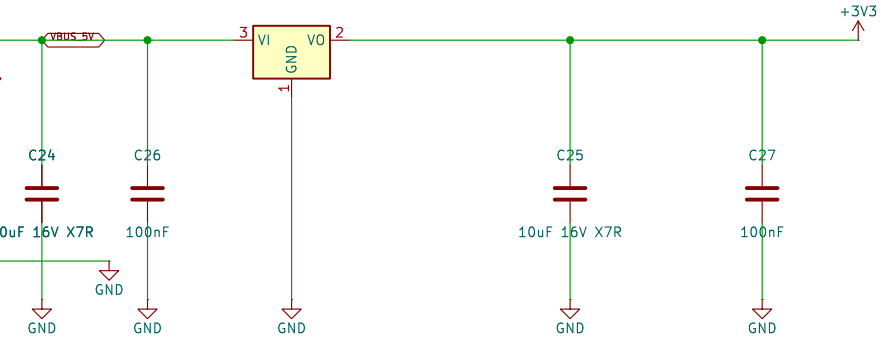




02 / 3.3 V linear regulator

AP7361C-33ER

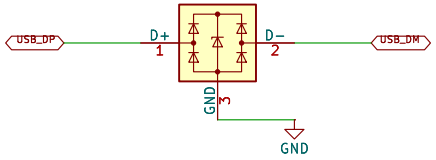
U22



03 / USB ESD protection

TPD2EUSB30ADRTR

U23



CERN-OHL-P-2.0

Sheet: /USB-C input and 3.3 V supply/
File: scp55-usb-power.kicad_sch

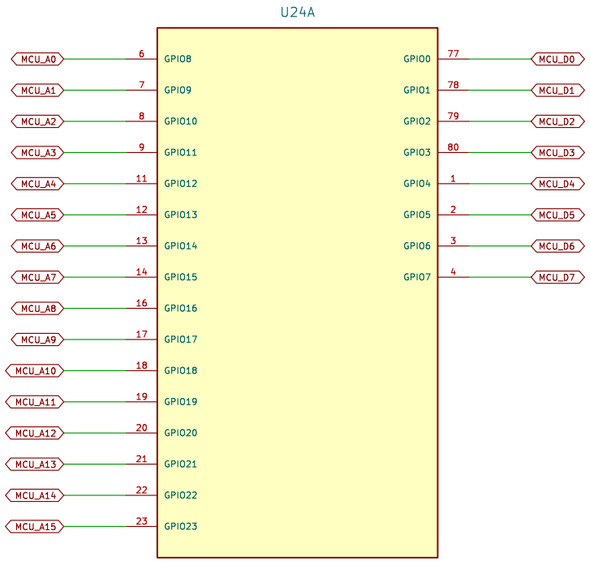
Title: USB-C input and 3.3 V supply

Size: A4	Date:	Rev: A / readable
KiCad E.D.A. 9.0.6		Id: 2/12

MCU / CARD BUS AND PERIPHERALS

U24 is one physical RP2354B package, drawn in four functional units (pages 3–4).

01 / Address and data bus



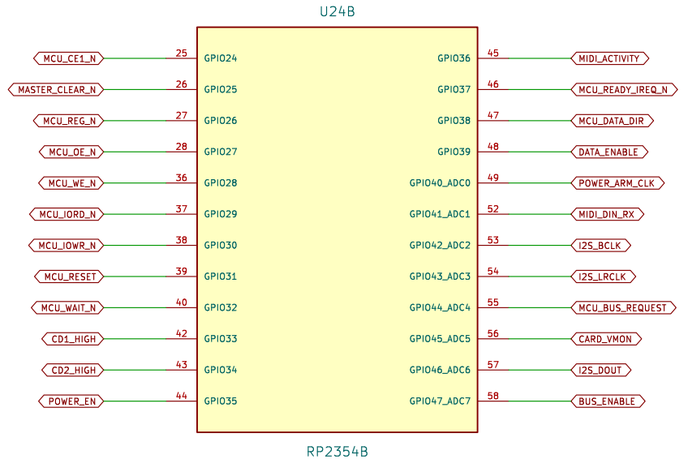
MCU_A[0..15] -> U2/U3 -> CARD_A[0..15] / page 5

MCU_D[0..7] <-> U1 <-> CARD_D[0..7] / page 5

POWER_ARM_CLK sets the latch; MASTER_CLEAR_N clears it.

CARD_VMON is half the card supply voltage. See page 8.

02 / Control, MIDI and I2S



MCU control -> U4/U6 -> card strobes / page 6

I2S -> PCM5102A DAC / page 11; DIN RX / page 10

CERN-OHL-P-2.0

Sheet: /MCU / card bus and peripherals/
File: scp55-mcu-signals.kicad_sch

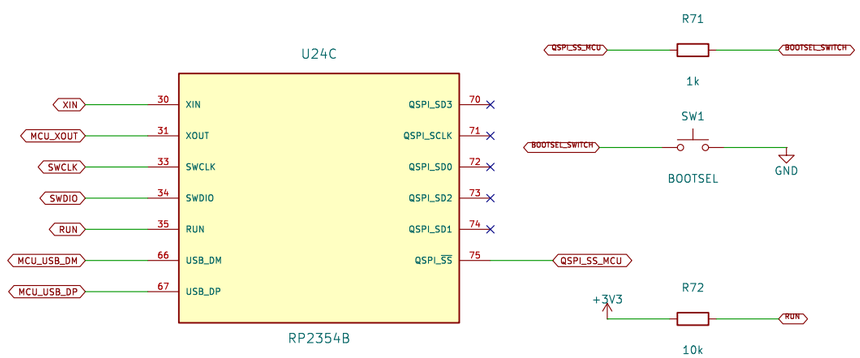
Title: MCU / card bus and peripherals

Size: A3	Date:	Rev: A / readable
KiCad E.D.A. 9.0.6		Id: 3/12

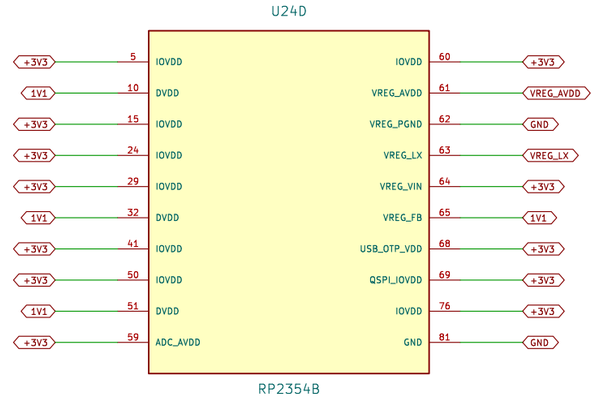
MCU / CLOCK, BOOT AND POWER

RP2354B includes 2 MB flash. QSPI data/clock connect internally; no external flash.

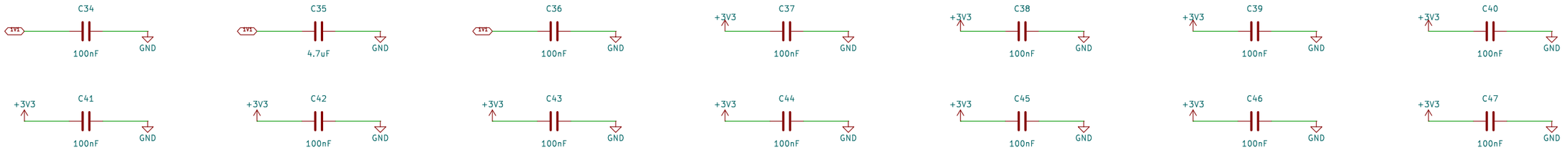
01 / Clock, USB, debug and boot



02 / Power pins and core regulator



03 / Local supply bypass



CERN-OHL-P-2.0

Sheet: /MCU / clock, boot and power/
File: scp55-mcu-support.kicad_sch

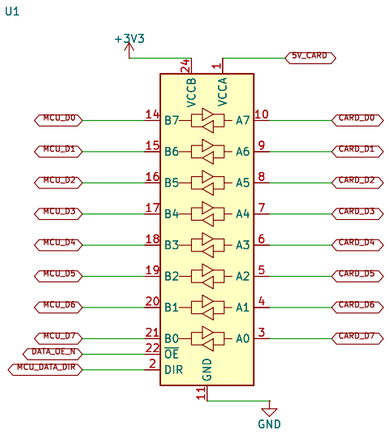
Title: MCU / clock, boot and power

Size: A3	Date:	**Rev: A / readable**
KiCad E.D.A. 9.0.6		Id: 4/12

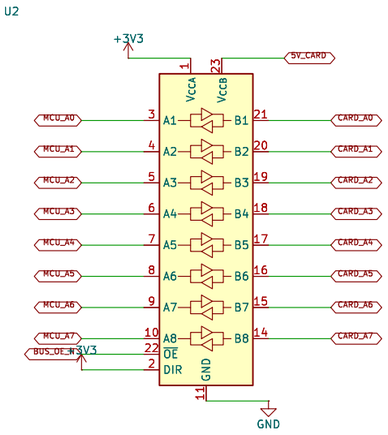
CARD BUS / ADDRESS AND DATA TRANSLATION

3.3 V MCU domain <–> 5 V card domain. Socket pinout is on page 7.

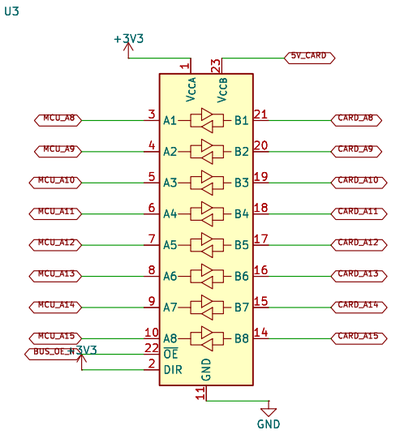
01 / Bidirectional data



02 / Address A0–A7



03 / Address A8–A15

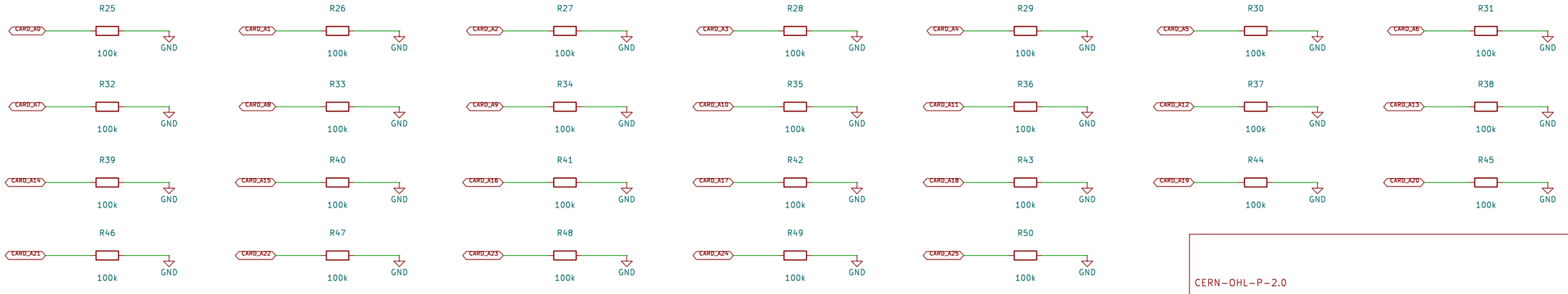








04 / Card–side idle pulldowns (100k)



CERN–OHL–P–2.0

Sheet: /Card bus / address and data translation/
File: scp55–bus–address–data.kicad_sch

Title: Card bus / address and data translation

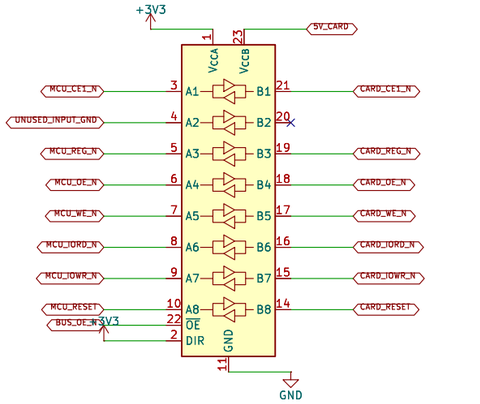
Size: A3	Date:	**Rev: A / readable**
KiCad E.D.A. 9.0.6		Id: 5/12

CARD BUS / CONTROL AND RETURN SIGNALS

Control strobes and RESET travel toward the card; status returns to the MCU.

01 / MCU -> card strobes and RESET

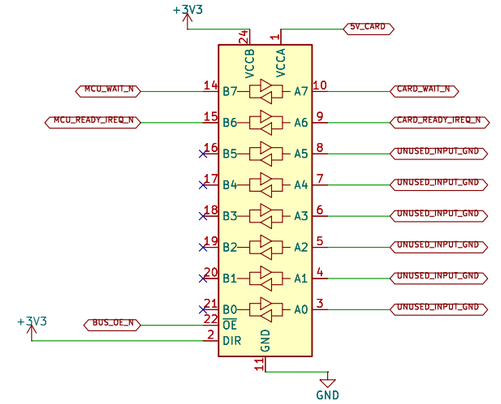
U4



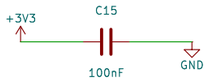
SN74LVC8T245PW

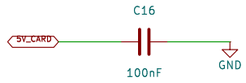
02 / Card -> MCU status

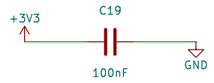
U6

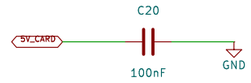


74LVC4245APW











0R VPP1 option



0R VPP2 option

BUS_ENABLE gates the control path.

U4 channel 8 carries the active-high card RESET.

U6 carries WAIT and READY/IREQ status back to the MCU.

Translator enable logic and removal latch: page 8.

CERN-OHL-P-2.0

Sheet: /Card bus / control and return signals/
File: scp55-bus-control.kicad_sch

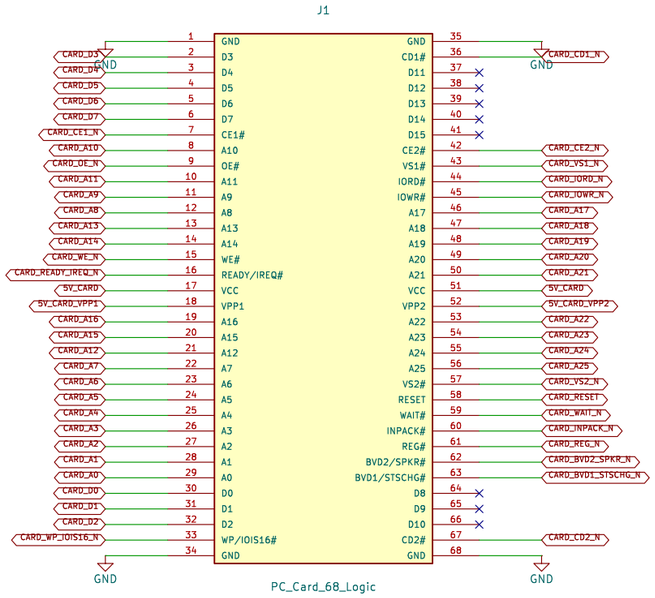
Title: Card bus / control and return signals

Size: A3	Date:	Rev: A / readable
KiCad E.D.A. 9.0.6		Id: 6/12

PC CARD SOCKET / PINOUT AND FIXED OPTIONS

J1 is the salvaged PC Card connector. Confirm physical pin numbering before assembly.

01 / SCP-55 PC Card connector

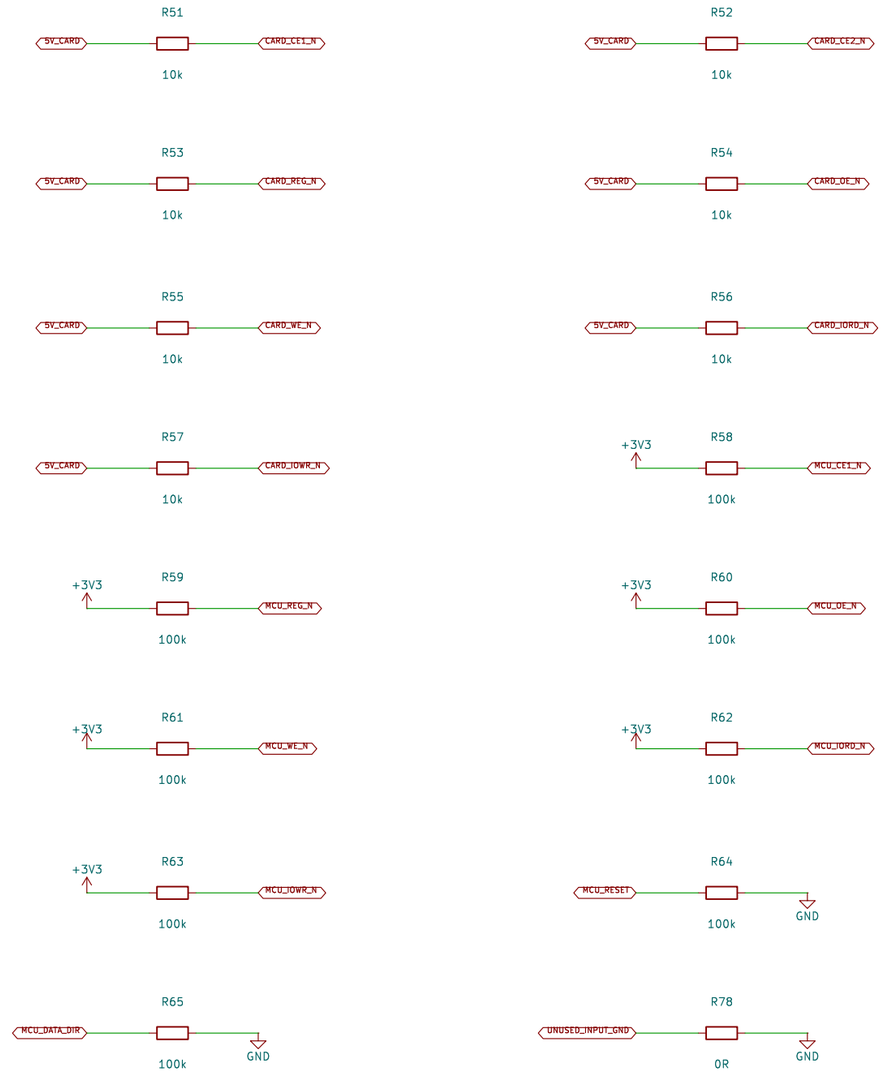


Address / data: page 5; control / status: page 6.

5 V card power and removal interlock: page 8.

CARD_CD1_N / CARD_CD2_N sensing: page 9.

02 / Card-side pull-ups and options



CERN-OHL-P-2.0

Sheet: /PC Card socket / pinout and fixed options/
File: scp55-card-socket.kicad_sch

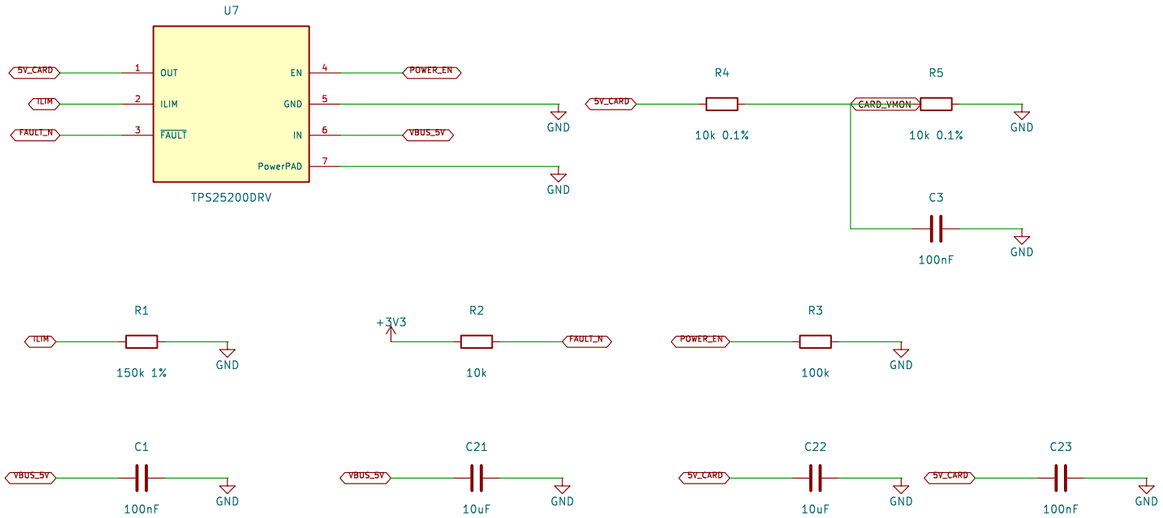
Title: PC Card socket / pinout and fixed options

Size: A3	Date:	Rev: A / readable
KiCad E.D.A. 9.0.6		Id: 7/12

CARD POWER / PROTECTION AND INTERLOCK

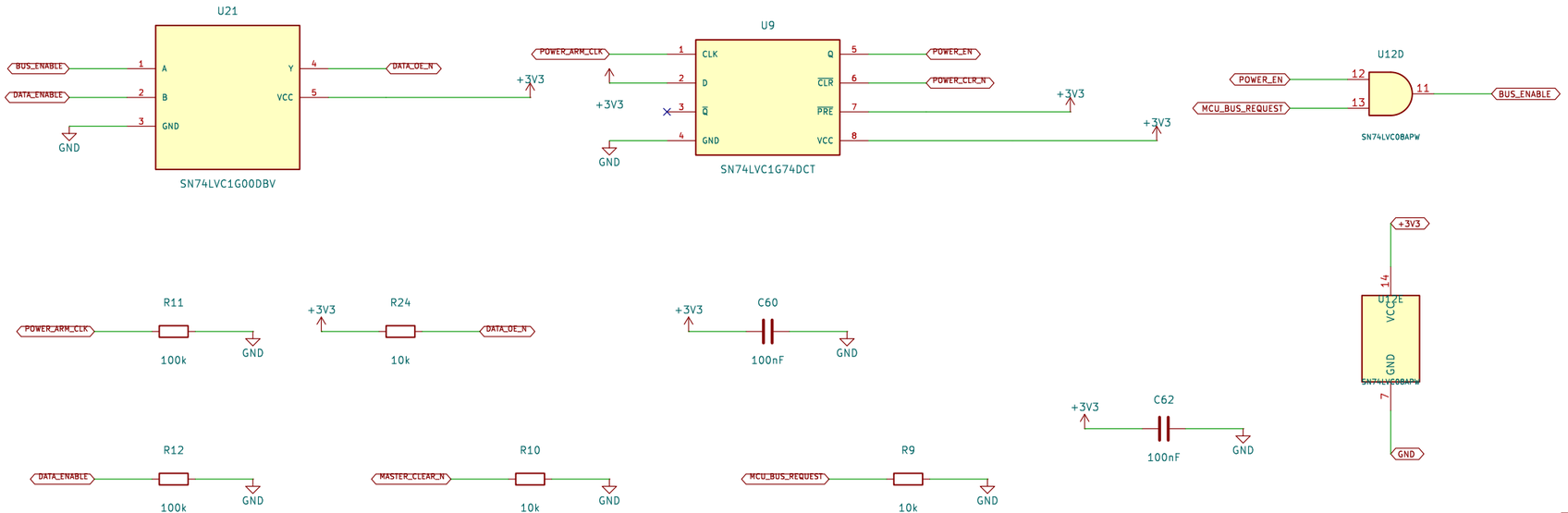
Removal or switch FAULT clears the latch. Contact restoration cannot restart the card.

01 / Switched card supply + ADC



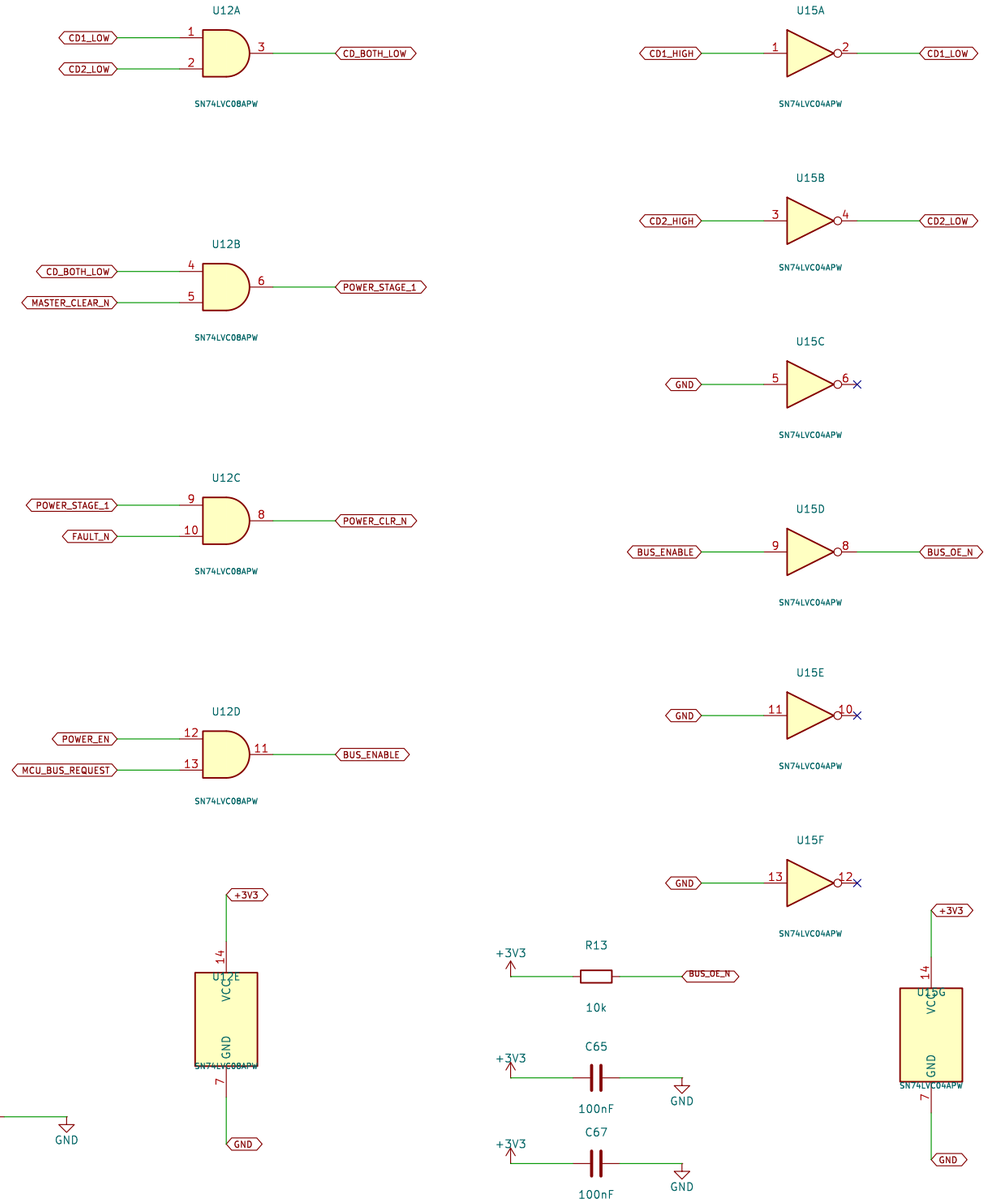
ADC divider: R4/R5 = 1/2. C3 filters CARD_VMON.

02 / Clear condition and power latch



Firmware: voltage check –> settle –> bus enable –> RESET.

03 / Enable qualification



CERN-OHL-P-2.0

Sheet: /Card power / protection and interlock/
File: scp55-card-control.kicad_sch

Title: Card power / protection and interlock

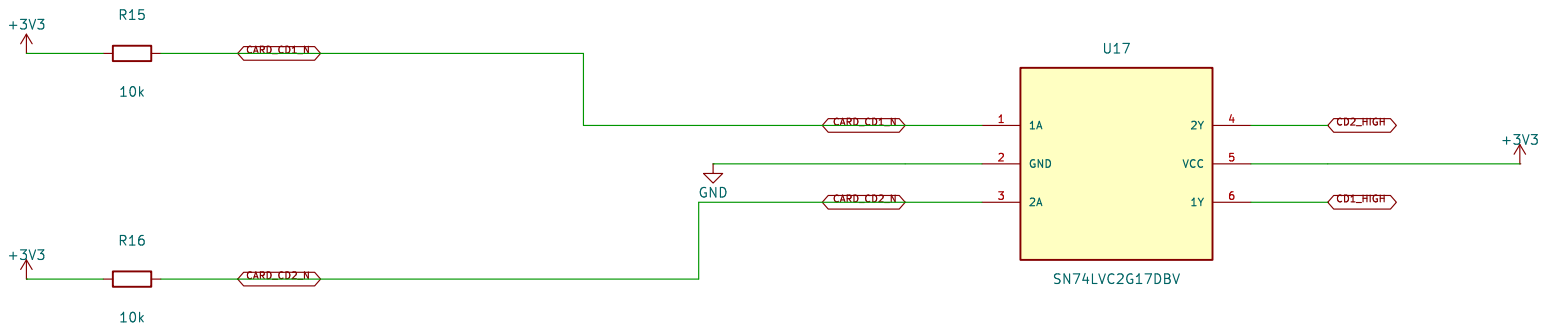
Size: A3	Date:	Rev: A / readable
KiCad E.D.A. 9.0.6		Id: 8/12

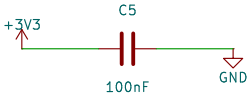
CARD DETECTION / SCHMITT INPUTS

SCP-55 only. Open CD1/CD2 contacts asynchronously shut down card power and the bus.

01 / Contacts and pull-ups

02 / Schmitt inputs





Inputs: J1 contact pins (page 7).

Outputs: MCU / interlock (pages 3, 8).

Both contacts required to arm power.

CERN-OHL-P-2.0

Sheet: /Card detection / Schmitt inputs/
File: scp55-card-detect.kicad_sch

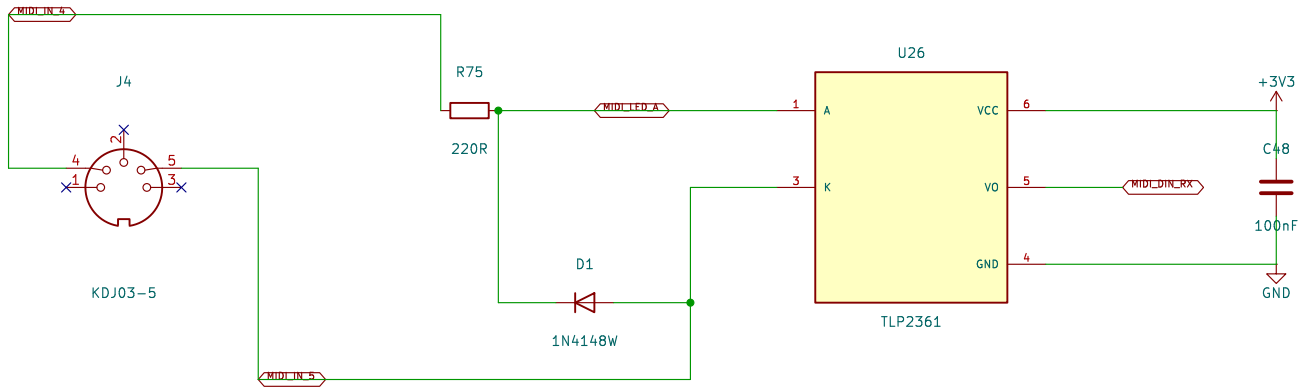
Title: Card detection / Schmitt inputs

Size: A4	Date:	Rev: A / readable
KiCad E.D.A. 9.0.6		Id: 9/12

DIN MIDI INPUT / ACTIVITY INDICATORS

Optically isolated MIDI input; USB MIDI enters through the MCU USB interface.

01 / DIN receiver -> UART1 RX





CERN-OHL-P-2.0

Sheet: /DIN MIDI input / activity indicators/
File: scp55-midi-in.kicad_sch

Title: DIN MIDI input / activity indicators

Size: A4	Date:	Rev: A / readable
KiCad E.D.A. 9.0.6		Id: 10/12

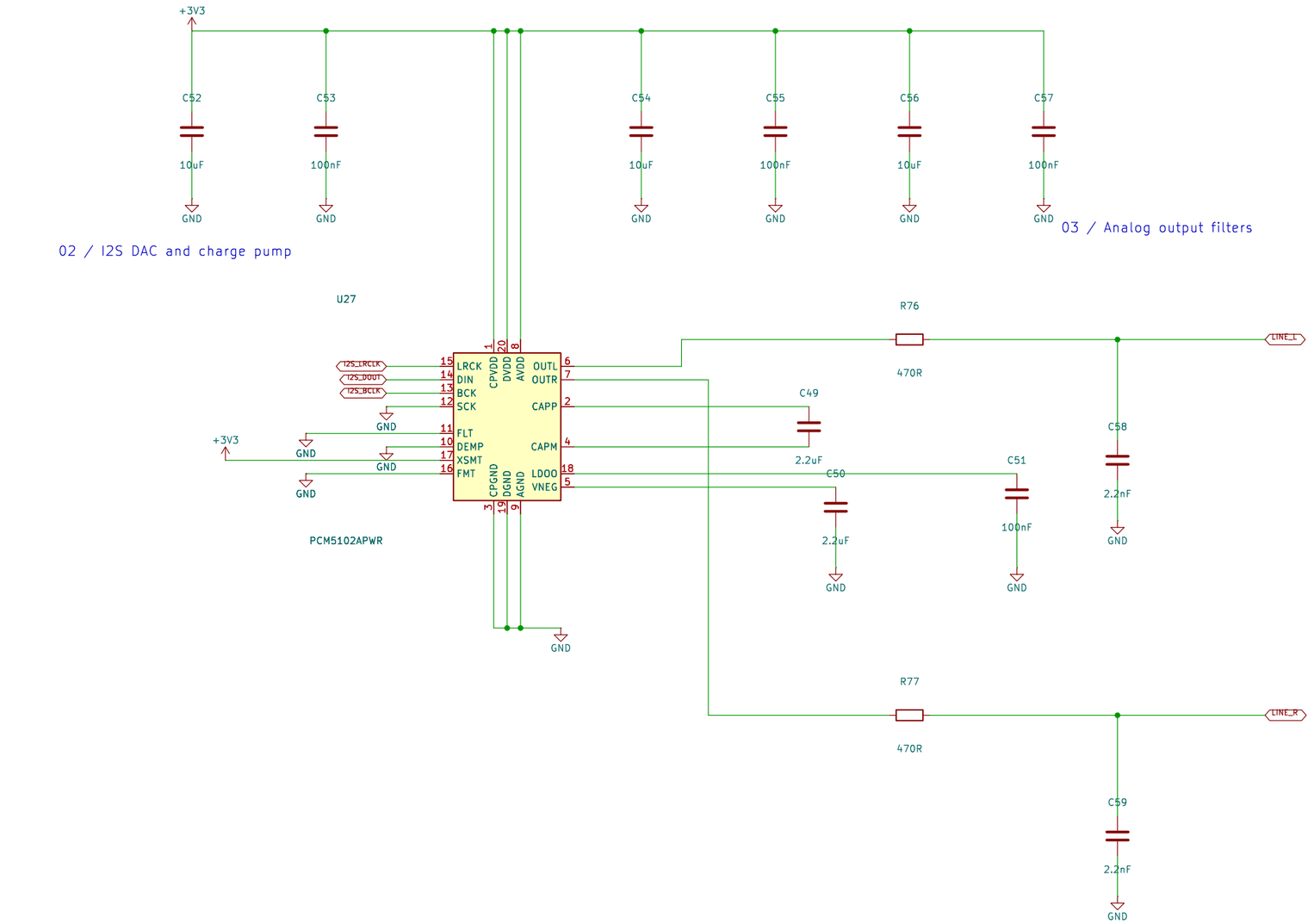
AUDIO OUTPUT / PCM5102A DAC

44.1 kHz / 16-bit stereo I2S from the MCU. LINE OUT level is set by external equipment.

01 / Supply bypass

02 / I2S DAC and charge pump

03 / Analog output filters



CERN-OHL-P-2.0

Sheet: /Audio output / PCM5102A DAC/
File: scp55-dac.kicad_sch

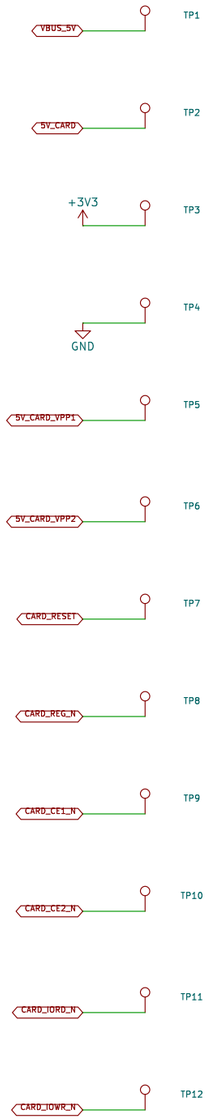
Title: Audio output / PCM5102A DAC

Size: A3	Date:	Rev: A / readable
KiCad E.D.A. 9.0.6		Id: 11/12

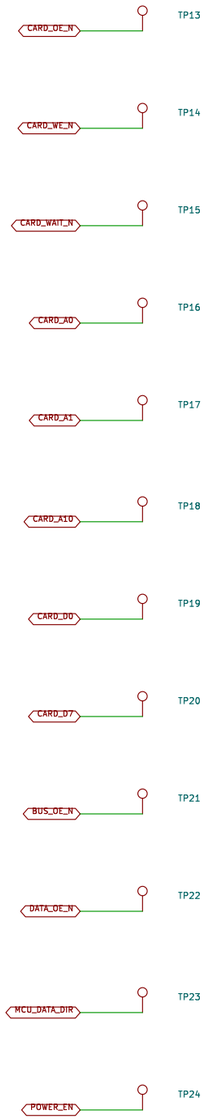
TEST POINTS / SIGNAL REFERENCE

Test-point numbers correspond to the production board; power flags are ERC-only symbols.

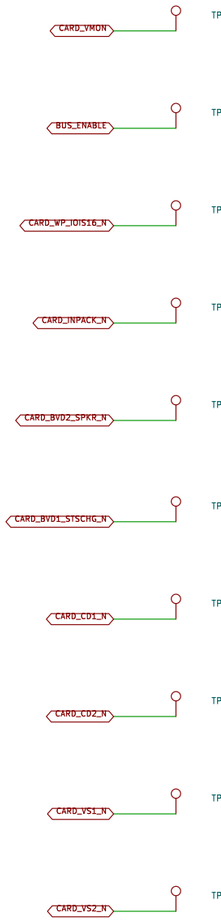
01 / Supply and card controls



02 / Card status and voltage



03 / Bus enables and MCU controls









CERN-OHL-P-2.0

Sheet: /Test points / signal reference/
File: scp55-debug.kicad_sch

Title: Test points / signal reference

Size: A3	Date:	**Rev: A / readable**
KiCad E.D.A. 9.0.6		Id: 12/12